CAMBRIDGE
International Education

Cambridge IGCSE™

DESIGN AND TECHNOLOGY **0445/11**

Paper 1 Product Design **May/June 2026**

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

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General marking principles

All examiners must apply these marking principles when marking candidate responses.

1	Examiners must award marks for each response in line with: • the specific **content** defined in the mark scheme • the specific **skills** defined in the mark scheme or in the level descriptions • the **standard of response** required as exemplified by the standardisation scripts.
2	Examiners must award **whole marks** (not half marks, or other fractions).
3	Examiners must award marks **positively**. This means that: • marks are not deducted for errors or omissions • marks are awarded for correct/valid answers as defined in the mark scheme and also for **other valid answers** which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate • the quality of spelling, punctuation and grammar are judged **only** when the mark scheme indicates that these features are **specifically assessed** in the question. However, the meaning of all responses must be unambiguous.
4	Examiners must **consistently** apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.
5	Examiners must use the **full range of marks** defined in the mark scheme, unless limited by the quality of the candidate responses seen.
6	Examiners must award marks according to the mark scheme and **must not** award marks with grade thresholds or grade descriptions in mind.

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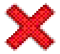
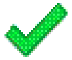
Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	Incorrect point
SEEN	Indicates that the point has been noted, but no credit has been given
	Correct point
Numbers	Indicating the mark allocated for the response

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Performance description tables

Each question contains some marks which are awarded using the following performance description tables.

Part (c)

Communication of ideas

Mark	Performance description
5–6	Ideas are communicated with precision and clarity through the use of accurate drawings and reasoned annotations linked to most of the requirements.
3–4	Ideas are displayed with some clarity through clear drawings supported by annotations referring to some of the requirements.
1–2	Simple drawings and limited annotations show little understanding of the requirements.
0	No creditable response.

Suitable designs

Mark	Performance description
5–6	Creative solutions which fully meet the requirements. Designs showing most aspects of construction detail.
3–4	Sensible solutions that mostly meet the requirements. Designs with moderate construction detail.
1–2	Solutions do not meet many of the requirements. Simplistic designs with little construction detail.
0	No creditable response

Part (e)

Quality of drawing

Mark	Performance description
4	High standard of line quality, use of colour and proportions. Appropriate techniques used that show clearly all detail.
2–3	Good line quality, use of colour and proportions. Most of the detail presented.
1	Poor line quality and proportions. Little detail presented.
0	No creditable response.

Construction details

Mark	Performance Description
5–6	All construction detail clear with good annotations and/or additional detail drawings as necessary.
3–4	Most construction may be obvious from overall views or with some annotation.
1–2	A simplistic design; little or no detail of construction used.
0	No creditable response.

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Guidance on using the performance description tables

Marking should be positive, rewarding achievement where possible but clearly differentiating across the whole range of marks available.
In approaching the assessment process, examiners should look at the work and then make a 'best fit' judgement as to which level statement it fits. In practice the work does not always match one level statement precisely so a judgement may need to be made between two or more level statements.

Once a 'best fit' level statement has been identified the following guide should be used to decide on a specific mark:

- Where the candidate's work **convincingly** meets the level statement, the highest mark should be awarded
- Where the candidate's work **adequately** meets the level statement, the most appropriate mark in the middle of the range should be awarded
- Where the candidate's work **just** meets the level statement, the lowest mark should be awarded

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Candidates answer **one** question, **either** 1 **or** 2 **or** 3.

Question	Answer	Marks	Guidance
1(a)	Accept any **four** additional specification points Must be waterproof, must go up to a set number e.g.10, size of numbers must be clear, font used must be clear to read, must be a stable structure on different ground types, must be easy to reach the score, inform about the score/teams. [1 × 4]	**4**	Each specification point – 1 mark No repeats from question: to be used on a sports pitch, display the score, display the score of two teams, use number boards, portable and allow scores to be regularly updated Only accept unqualified answers if relevant to this specific design problem, **not** generic answers such as nice, safe, aesthetic Acceptable one word answers: lightweight, weatherproof/waterproof, rustproof, durable, stable, foldable, stable or sturdy Any other valid response
1(b)	Accept drawings of any **two** methods of temporarily displaying items: cards on hooks, rotational dial, pop-up numbers on cards, chalk board for writing numbers. [2 × 2]	**4**	Maximum of 2 marks for each drawing: Appropriate temporary display method – 1 mark Clear drawing – 1 mark Any other valid response
1(c)	Any **three** suitable ideas. Award up to **6 marks for communication of ideas** using the ‘Communication of ideas’ table. Award up to **6 marks for suitable designs** using the ‘Suitable designs’ table.	**12**	At least **three different** ideas for maximum marks. Pro rata if fewer.

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Question	Answer	Marks	Guidance
1(d)	Award up to **6 marks for evaluation** of the ideas: Evaluation [2 × 3] e.g. Advantage + disadvantage explained for each idea Selection [1] Justification – **not** single words, or generic terms such as the best, meets the specification or most suitable [1]	**8**	Simple descriptions or repeats of same points for each idea not rewarded. Specific not generic justification. Award maximum marks if only either advantage or disadvantage given for each as long as includes sophisticated reasoning.
1(e)	Award up to **4 marks for quality of drawing** using the 'Quality of drawing' table. Award up to **2 marks for dimensions:** 2 or 3 overall dimensions only – **1 mark** Additional detail dimensions – **1 mark** Award up to **6 marks for construction detail** using the 'Construction details' table.	**12**	Additional detail dimensions might show thickness of materials, diameters, etc.
1(f)	Accept any **two** suitable **specific** materials. [1 × 2] Accept any **appropriate** reason for choice of **each** material [1 × 2]	**4**	Each suitable specific material – 1 mark Generic terms such as wood, metal, plastic **not** accepted. Appropriate reason for each material – 1 mark Materials must be appropriate for the design shown in **(e)**
1(g)	Accept any suitable manufacturing process. [1 × 1]	**1**	Process must be appropriate for design in **(e)**.
	Award up to **3 marks for description of process**.	**3**	Detailed description for 3 marks
	Award up to **2 marks for names of tools, equipment or machines used**.	**2**	Basic marking out tools, such as pencil or rule, or just drawings of tools/equipment = 1 mark only

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Question	Answer	Marks	Guidance
OR			
2(a)	Accept any **four** additional specification points: must be a stable piece of packaging, must have a viewing window to see the contents, must be easy to open the packaging, must advertise the contents, must have a flat base [1 × 4]	**4**	Each specification point – 1 mark No repeats from question: for a sports company, for three balls/an outdoor game, securely hold three different size balls, can be stacked on a shelf, balls need to be seen and references to the given sizes Only accept unqualified answers if relevant to this specific design problem, **not** generic answers such as safe, strong, nice, portable, stackable Acceptable one word answers: lightweight, weatherproof or waterproof (not both), durable, recyclable, aesthetic. Any other valid response
2(b)	Accept drawings of any **two** methods of holding a ball in place only using lightweight materials: net using holes to wrap around sphere, sticky pads, zip ties going through card surround, vacuum formed PVC [2 × 2]	**4**	Maximum of 2 marks for each drawing: Method – 1 mark Clear drawing – 1 mark Any other valid response
2(c)	Any **three** suitable ideas. Award up to **6 marks for communication of ideas** using the ‘Communication of ideas’ table. Award up to **6 marks for suitable designs** using the ‘Suitable designs’ table.	**12**	At least **three different** ideas for maximum marks. Pro rata if fewer.

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Question	Answer	Marks	Guidance
2(d)	Award up to **6 marks for evaluation** of the ideas: Evaluation [2 × 3] e.g. Advantage + disadvantage explained for each idea Selection [1] Justification – **not** single words, or generic terms such as the best, meets the specification or most suitable [1]	**8**	Simple descriptions or repeats of same points for each idea not rewarded. Specific not generic justification. Award maximum marks if only either advantage or disadvantage given for each as long as includes sophisticated reasoning.
2(e)	Award up to **4 marks for quality of drawing** using the 'Quality of drawing' table. Award up to **2 marks for dimensions:** 2 or 3 overall dimensions only – **1 mark** Additional detail dimensions – **1 mark** Award up to **6 marks for construction detail** using the 'Construction details' table.	**12**	Additional detail dimensions might show thickness of materials, diameters, etc.
2(f)	Accept any **two** suitable **specific** materials. [1 × 2] Accept any **appropriate** reason for choice of **each** material [1 × 2]	**4**	Each suitable specific material – 1 mark Generic terms such as wood, metal, plastic **not** accepted. Appropriate reason for each material – 1 mark Materials must be appropriate for the design shown in **(e)**
2(g)	Accept any suitable manufacturing process. [1 × 1].	**1**	Process must be appropriate for design in **(e)**.
	Award up to **3 marks for description of process**.	**3**	Detailed description for 3 marks
	Award up to **2 marks for names of tools, equipment or machines used.**	**2**	Basic marking out tools, such as pencil or rule, or just drawings of tools/equipment = 1 mark only

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Question	Answer	Marks	Guidance
OR			
3(a)	Accept any **four** additional specification points: must be easy to reload, trigger and handle must be ergonomic, device must be safe to use, must use a simple mechanism so it is easy to understand, durable mechanism, reliable mechanism, material/mechanism that is safe for the user (children) [1 × 4]	**4**	Each specification point – 1 mark No repeats from question: used outdoors, uses a target, safely propel a foam ball at a target, handheld, uses a trigger to release the ball and references to the given ball size Only accept unqualified answers if relevant to this specific design problem, **not** safe, nice. Acceptable one word answers: lightweight, ergonomic, weatherproof or waterproof (not both), durable, accurate, aesthetic. Any other valid response
3(b)	Accept drawings of any **two** methods for propelling an object in a linear direction, spring release mechanism, rubber band powered, air pump, lever action. [2 × 2]	**4**	Maximum of 2 marks for each drawing: Method – 1 mark Clear drawing – 1 mark Any other valid response
3(c)	Any **three** suitable ideas. Award up to **6 marks for communication of ideas** using the 'Communication of ideas' table. Award up to **6 marks for suitable designs** using the 'Suitable designs' table.	**12**	At least **three different** ideas for maximum marks. Pro rata if fewer.

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Question	Answer	Marks	Guidance
3(d)	Award up to **6 marks for evaluation** of the ideas: Evaluation [2 × 3] e.g. Advantage + disadvantage explained for each idea Selection [1] Justification – **not** single words, or generic terms such as the best, meets the specification or most suitable [1]	**8**	Simple descriptions or repeats of same points for each idea not rewarded. Specific not generic justification. Award maximum marks if only either advantage or disadvantage given for each as long as includes sophisticated reasoning.
3(e)	Award up to **4 marks for quality of drawing** using the 'Quality of drawing' table. Award up to **2 marks for dimensions:** 2 or 3 overall dimensions only – **1 mark** Additional detail dimensions – **1 mark** Award up to **6 marks for construction detail** using the 'Construction details' table.	**12**	Additional detail dimensions might show thickness of materials, diameters, etc.
3(f)	Accept any **two** suitable **specific** materials. [1 × 2] Accept any **appropriate** reason for choice of **each** material [1 × 2]	**4**	Each suitable specific material – 1 mark Generic terms such as wood, metal, plastic **not** accepted. Appropriate reason for each material – 1 mark Materials must be appropriate for the design shown in **(e)**
3(g)	Accept any suitable manufacturing process. [1 × 1]	**1**	Process must be appropriate for design in **(e)**.
	Award up to **3 marks for description of process**.	**3**	Detailed description for 3 marks
	Award up to **2 marks for names of tools, equipment or machines used**.	**2**	Basic marking out tools, such as pencil or rule, or just drawings of tools/equipment = 1 mark only

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